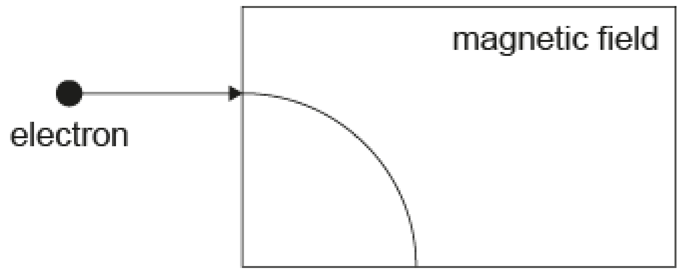
Study Guide - SL - Topic B5 D1 D2 D3 - Paper 1A [41 marks]

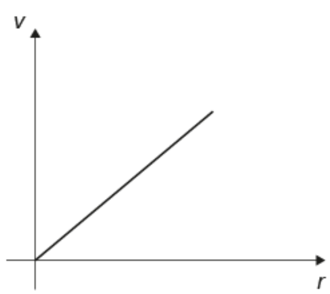
1. [Maximum mark: 1] 23M.1.SL.TZ1.21

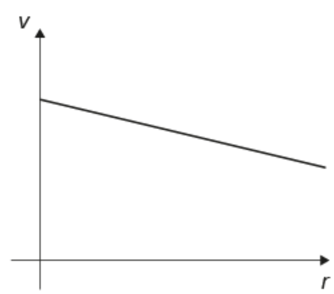
An electron enters a region of uniform magnetic field at a speed v. The direction of the electron is perpendicular to the magnetic field. The path of the electron inside the magnetic field is circular with radius r.

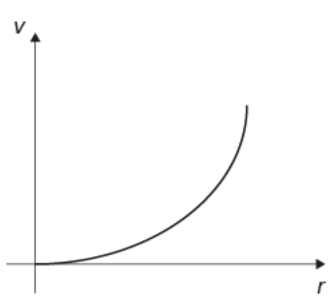


The speed of the electron is varied to obtain different values of r.

Which graph represents the variation of speed v with r?

A.


B.


C.


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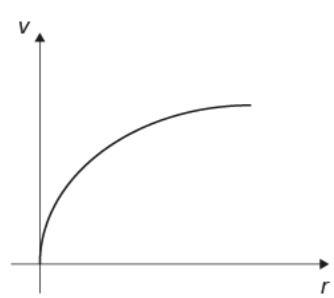


[1]

Markscheme

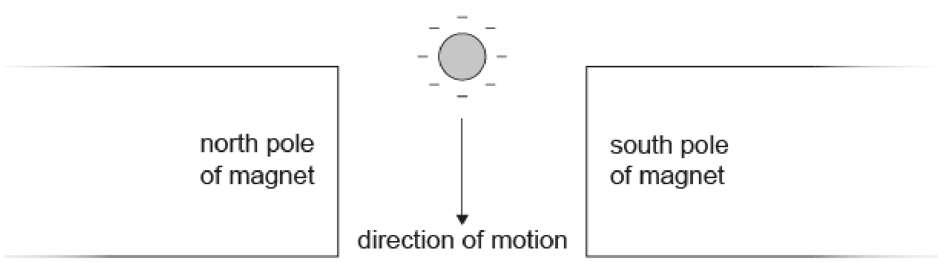
A

Examiners report

Option A was the (correct) answer chosen most frequently by HL students, although option D was a significant distractor. More SL students selected option D than A, perhaps confusing the graph of speed vs r with that of acceleration vs *r*.

2. [Maximum mark: 1] 23M.1.SL.TZ2.21

A negatively charged sphere is falling through a magnetic field.



What is the direction of the magnetic force acting on the sphere?

A. To the left of the page

B. To the right of the page

C. Out of the page

D. Into the page [1]

Markscheme

D

3. [Maximum mark: 1] 23M.1.SL.TZ2.22

An electron is accelerated from rest through a potential difference *V*.

What is the maximum speed of the electron?

A. $\sqrt{\frac{2eV}{m_e}}$

B. $\frac{eV}{m_e}$

C. $\frac{2eV}{m_e}$

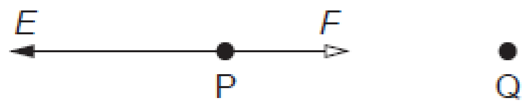
D. $\sqrt{\frac{2V}{m_e}}$

[1]

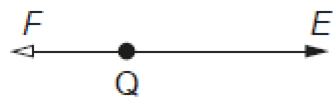
Markscheme
A

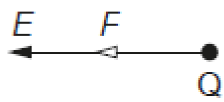
4. [Maximum mark: 1] 22M.1.SL.TZ1.19

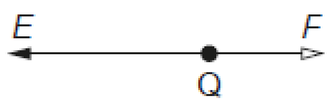
P and Q are two opposite point charges. The force F acting on P due to Q and the electric field strength E at P are shown.

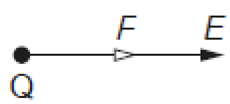


Which diagram shows the force on Q due to P and the electric field strength at Q?

A. 

B. 

C. 

D. 

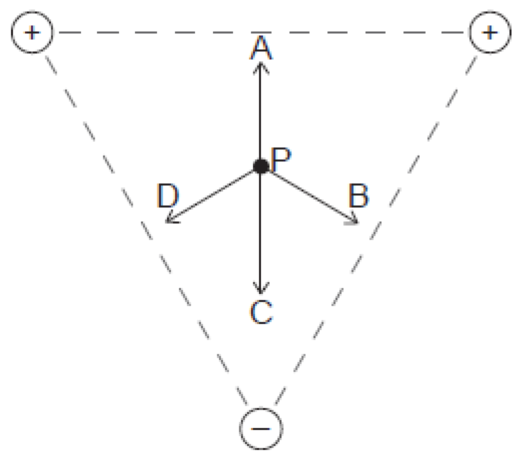
[1]

Markscheme
B

Examiners report
Option A was the most frequent answer selected by both HL and SL candidates, suggesting that determining the direction of the electric field was more problematic than the direction of the force (Newton 3).

5. [Maximum mark: 1] 22M.1.SL.TZ1.20

Three point charges of equal magnitude are placed at the vertices of an equilateral triangle. The signs of the charges are shown. Point P is equidistant from the vertices of the triangle. What is the direction of the resultant electric field at P?



[1]

Markscheme

C

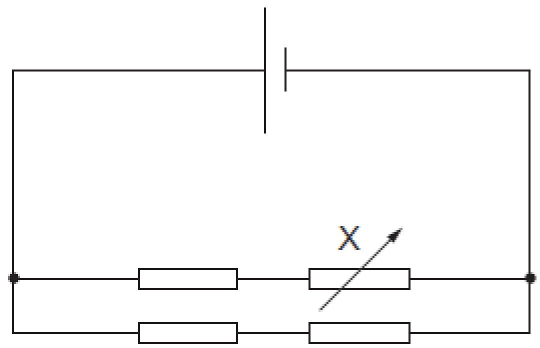
Examiners report

This question was very well answered by both HL and SL candidates, reflected in the high difficulty level for both papers.

6. [Maximum mark: 1] 22M.1.SL.TZ1.21

Three identical resistors each of resistance *R* are connected with a variable resistor X as shown. X is initially set to *R*. The current in the cell is 0.60 A.

The cell has negligible internal resistance.



X is now set to zero. What is the current in the cell?

A. 0.45 A

B. 0.60 A

C. 0.90 A

D. 1.80 A [1]

Markscheme

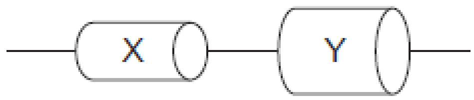
C

Examiners report

Option C was the most common (correct) answer, however option B was also a frequent response. This question had a relatively high discrimination index, suggesting that more able candidates had less difficulty managing resistance in this combination circuit.

7. [Maximum mark: 1] 22M.1.SL.TZ1.22

Two cylinders, X and Y, made from the same material, are connected in series.



The cross-sectional area of Y is twice that of X. The drift speed of the electrons in X is v_X and in Y it is v_Y.

What is the ratio $\frac{v_X}{v_Y}$?

A. 4

B. 2

C. 1

D. $\frac{1}{2}$ [1]

Markscheme

B

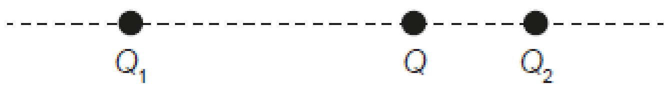
Examiners report

Responses from SL candidates were split between options B and D. Candidates appeared to recognize the factor of two in the drift speed ratio, but were unclear as to whether it was 2:1 or 1:2. Candidates are encouraged to think if their answer makes sense given the context of the question; it is likely that common sense would help candidates in this instance. Practice manipulating ratios to compare changing variables is a useful skill, and this question could be put to good use in this regard.

8. [Maximum mark: 1] 22M.1.SL.TZ2.19

A charge Q is at a point between two electric charges Q_1 and Q_2. The net electric force on Q is zero. Charge Q_1 is further from Q than charge Q_2.

What is true about the signs of the charges Q_1 and Q_2 and their magnitudes?



	Signs of charges Q_1 and Q_2	Magnitudes
A.	same	$Q_1 > Q_2$
B.	same	$Q_1 < Q_2$
C.	opposite	$Q_1 > Q_2$
D.	opposite	$Q_1 < Q_2$

[1]

Markscheme

A

9. [Maximum mark: 1] 22M.1.SL.TZ2.20

A battery of negligible internal resistance is connected to a lamp. A second identical lamp is added in series. What is the change in potential difference across the first lamp and what is the change in the output power of the battery?

	Change in potential difference	Output power of battery
A.	decreases	decreases
B.	decreases	increases
C.	no change	decreases
D.	no change	increases

[1]

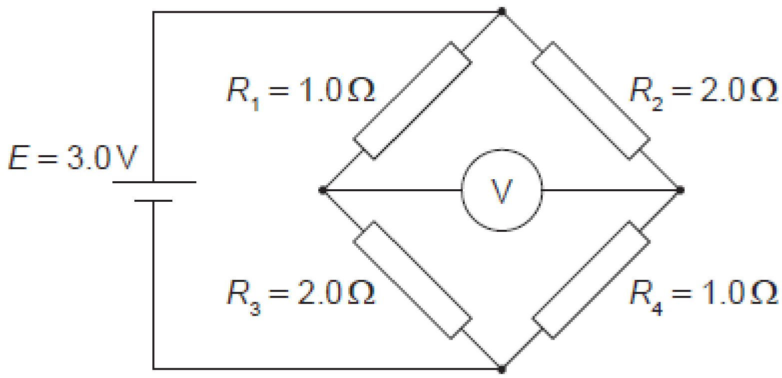
Markscheme

A

10. [Maximum mark: 1] 22M.1.SL.TZ2.21

A circuit consists of a cell of emf $E = 3.0$ V and four resistors connected as shown. Resistors R_1 and R_4 are 1.0 Ω and resistors R_2 and R_3 are 2.0 Ω.

What is the voltmeter reading?



A. 0.50 V

B. 1.0 V

C. 1.5 V

D. 2.0 V [1]

Markscheme

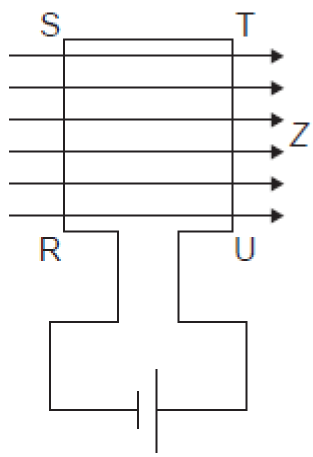
B

Examiners report

There were some comments from teachers that the circuit is unfamiliar, however it is basically a series and parallel circuit and can be solved by considering the parallel sections individually either by calculating the current through each and then the voltages across the individual resistors or by considering the resistors as a potential divider. It has a low discrimination index at HL with many choosing option C (B correct) and very poor discrimination at SL, again with option C the most popular choice.

11. [Maximum mark: 1] 22M.1.SL.TZ2.22

A rectangular coil of wire RSTU is connected to a battery and placed in a magnetic field *Z* directed to the right. Both the plane of the coil and the magnetic field direction are in the same plane.



What is true about the magnetic force acting on the sides RS and ST?

	Force acting on RS	Force acting on ST
A.	into the page	into the page
B.	out of the page	no force acts
C.	into the page	no force acts
D.	out of the page	out of the page

[1]

Markscheme

B

12. [Maximum mark: 1] 22M.1.SL.TZ2.24

Three statements about Newton's law of gravitation are:

I. It can be used to predict the motion of a satellite.
II. It explains why gravity exists.
III. It is used to derive the expression for gravitational potential energy.

Which combination of statements is correct?

A. I and II only

B. I and III only

C. II and III only

D. I, II and III [1]

Markscheme

B

Examiners report

Comments suggested that 'gravitational potential' is more suitable for an HL question. However, candidates should have realised that statement II is incorrect so option B is the only possibility and this proved the most popular answer. The wording will be altered to 'gravitational potential energy' for publication.

13. [Maximum mark: 1] 21M.1.SL.TZ1.18

Two charges Q_1 and Q_2, each equal to 2 nC, are separated by a distance 3 m in a vacuum. What is the electric force on Q_2 and the electric field due to Q_1 at the position of Q_2?

	Electric force on Q_2	Electric field due to Q_1 at the position of Q_2
A.	4×10^{-9} N	2 N C^{-1}
B.	4 N	2 N C^{-1}
C.	4×10^{-9} N	2×10^{-9} N C^{-1}
D.	4 N	2×10^{-9} N C^{-1}

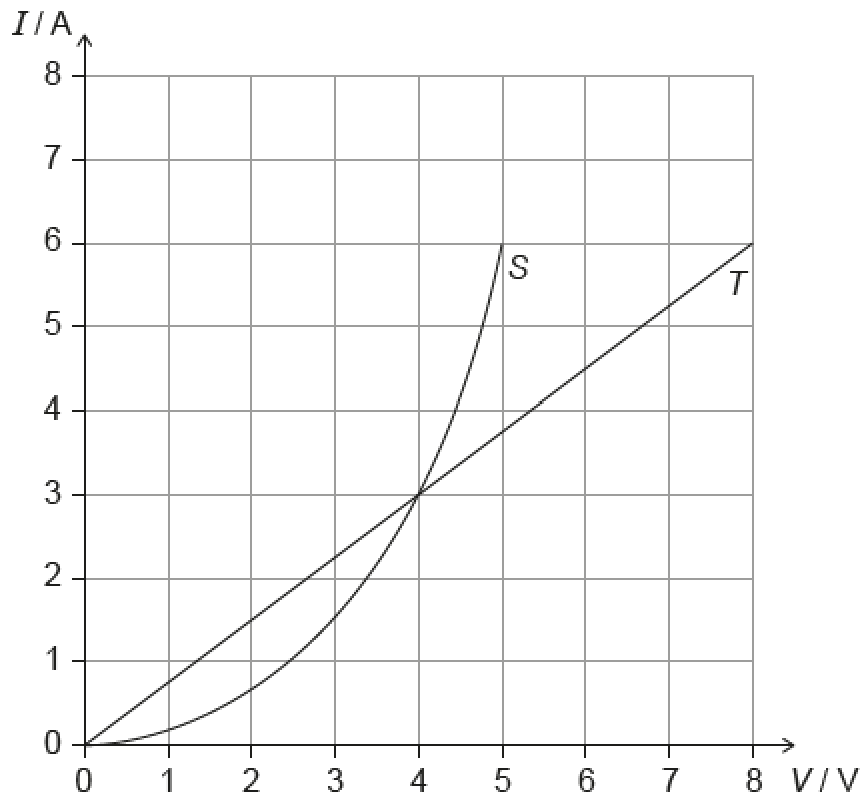
[1]

Markscheme

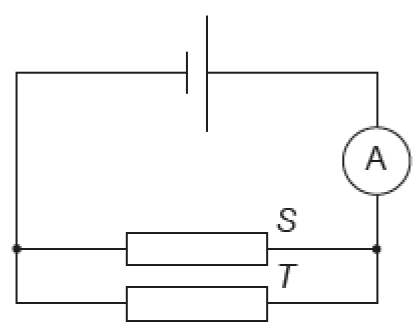
A

14. [Maximum mark: 1] 21M.1.SL.TZ1.19

Two conductors *S* and *T* have the *V*/*I* characteristic graphs shown below.



When the conductors are placed in the circuit below, the reading of the ammeter is 6.0 A.



What is the emf of the cell?

A. 4.0 V

B. 5.0 V

C. 8.0 V

D. 13 V

[1]

Markscheme
A

15. [Maximum mark: 1] 21M.1.SL.TZ1.20

For a real cell in a circuit, the terminal potential difference is at its closest to the emf when

A. the internal resistance is much smaller than the load resistance.

B. a large current flows in the circuit.

C. the cell is not completely discharged.

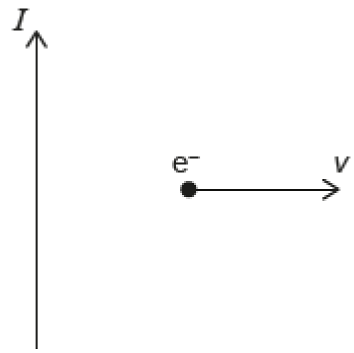
D. the cell is being recharged.

[1]

Markscheme
A

16. [Maximum mark: 1] 21M.1.SL.TZ1.21

A long straight vertical conductor carries a current I upwards. An electron moves with horizontal speed v to the right.



What is the direction of the magnetic force on the electron?

A. Downwards

B. Upwards

C. Into the page

D. Out of the page [1]

Markscheme
A

17. [Maximum mark: 1] 21M.1.SL.TZ1.23

Which is the definition of gravitational field strength at a point?

A. The sum of the gravitational fields created by all masses around the point

B. The gravitational force per unit mass experienced by a small point mass at that point

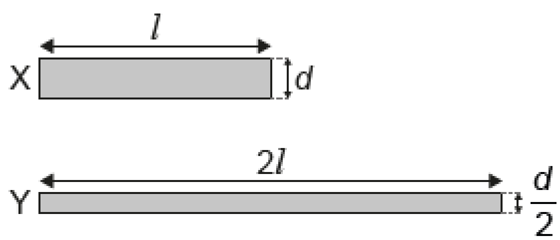
C. $G\frac{M}{r^2}$, where M is the mass of a planet and r is the distance from the planet to the point

D. The resultant force of gravitational attraction on a mass at that point [1]

Markscheme
B

18. [Maximum mark: 1] 21M.1.SL.TZ2.18

The diagram shows two cylindrical wires, X and Y. Wire X has a length l, a diameter d, and a resistivity ρ. Wire Y has a length $2l$, a diameter of $\frac{d}{2}$ and a resistivity of $\frac{\rho}{2}$.



What is $\frac{\text{resistance of X}}{\text{resistance of Y}}$?

A. 4

B. 2

C. 0.5

D. 0.25 [1]

Markscheme
D

19. [Maximum mark: 1] 21M.1.SL.TZ2.19

An ion moves in a circle in a uniform magnetic field. Which single change would increase the radius of the circular path?

A. Decreasing the speed of the ion

B. Increasing the charge of the ion

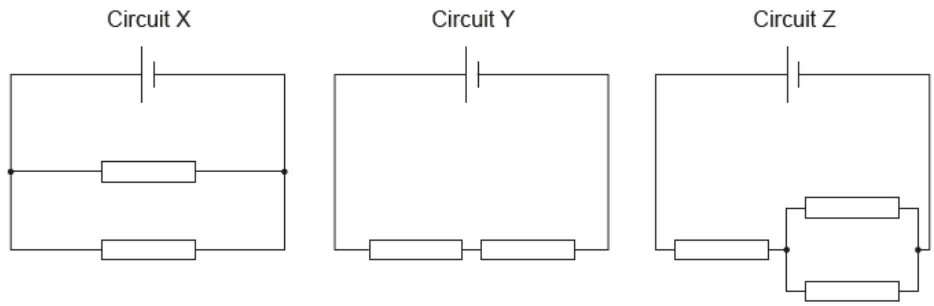
C. Increasing the mass of the ion

D. Increasing the strength of the magnetic field [1]

Markscheme
C

20. [Maximum mark: 1] 21M.1.SL.TZ2.20

In the circuits shown, the cells have the same emf and zero internal resistance. All resistors are identical.



What is the order of increasing power dissipated in each circuit?

	Lowest → Highest power dissipated
A.	Y, Z, X
B.	X, Y, Z
C.	X, Z, Y
D.	Y, X, Z

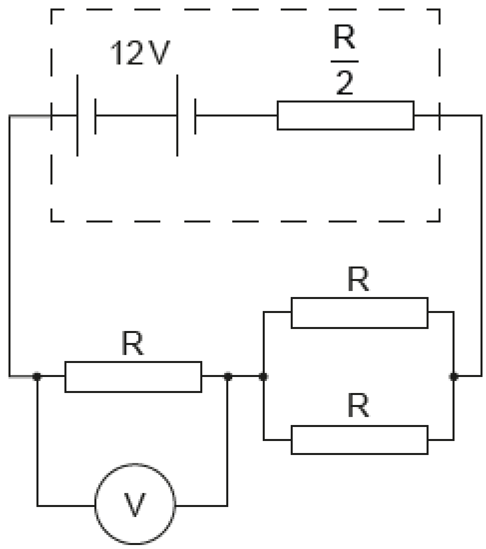
[1]

Markscheme

A

21. [Maximum mark: 1] 21M.1.SL.TZ2.21

Three identical resistors of resistance R are connected as shown to a battery with a potential difference of $12\,\mathrm{V}$ and an internal resistance of $\frac{\mathrm{R}}{2}$. A voltmeter is connected across one of the resistors.



What is the reading on the voltmeter?

A. $3\,\mathrm{V}$

B. $4\,\mathrm{V}$

C. $6\,\mathrm{V}$

D. $8\,\mathrm{V}$ [1]

Markscheme

C

22. [Maximum mark: 1] 21M.1.SL.TZ2.22

Magnetic field lines are an example of

A. a discovery that helps us understand magnetism.

B. a model to aid in visualization.

C. a pattern in data from experiments.

D. a theory to explain concepts in magnetism.

[1]

Markscheme
B

23. [Maximum mark: 1] 19M.1.SL.TZ1.21

Two cells each of emf 9.0 V and internal resistance 3.0 Ω are connected in series. A 12.0 Ω resistor is connected in series to the cells. What is the current in the resistor?

A. 0.50 A

B. 0.75 A

C. 1.0 A

D. 1.5 A [1]

Markscheme
C

Examiners report
This question was well answered by candidates and had a higher discrimination index.

24. [Maximum mark: 1] 19M.1.SL.TZ1.22

Charge flows through a liquid. The charge flow is made up of positive and negative ions. In one second 0.10 C of negative ions flow in one direction and 0.10 C of positive ions flow in the opposite direction.

What is the magnitude of the electric current flowing through the liquid?

A. 0 A

B. 0.05 A

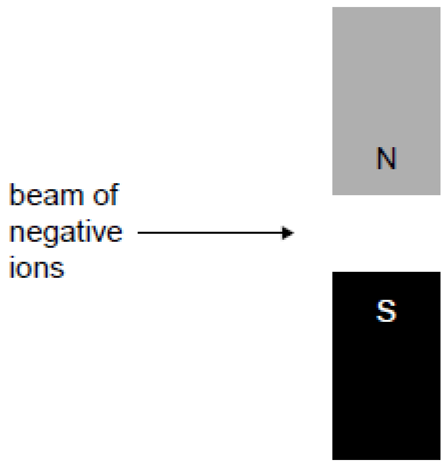
C. 0.10 A

D. 0.20 A [1]

Markscheme
D

25. [Maximum mark: 1] 19M.1.SL.TZ1.23

A beam of negative ions flows in the plane of the page through the magnetic field due to two bar magnets.



What is the direction in which the negative ions will be deflected?

A. Out of the page ⊙

B. Into the page X

C. Up the page ↑

D. Down the page ↓ [1]

Markscheme
A

Examiners report
This question was well answered by candidates.

26. [Maximum mark: 1] 19M.1.SL.TZ1.25

Satellite X orbits a planet with orbital radius *R*. Satellite Y orbits the same planet with orbital radius 2*R*. Satellites X and Y have the same mass.

What is the ratio $\frac{\text{centripetal acceleration of X}}{\text{centripetal acceleration of Y}}$?

A. $\frac{1}{4}$

B. $\frac{1}{2}$

C. 2

D. 4 [1]

Markscheme

D

27. [Maximum mark: 1] 19M.1.SL.TZ2.2

What is the unit of electrical potential difference expressed in fundamental SI units?

A. $kg\ m\ s^{-1}\ C^{-1}$

B. $kg\ m^{2}\ s^{-2}\ C^{-1}$

C. $kg\ m^{2}\ s^{-3}\ A^{-1}$

D. $kg\ m^{2}\ s^{-1}\ A$ [1]

Markscheme

C

Examiners report

The most popular answer was B giving a low discrimination index for this question. It should be a relatively straightforward question provided the candidate can remember which of 'C' or 'A' is the fundamental unit.

28. [Maximum mark: 1] 19M.1.SL.TZ2.18

A particle with a charge *ne* is accelerated through a potential difference *V*.

What is the magnitude of the work done on the particle?

A. eV

B. neV

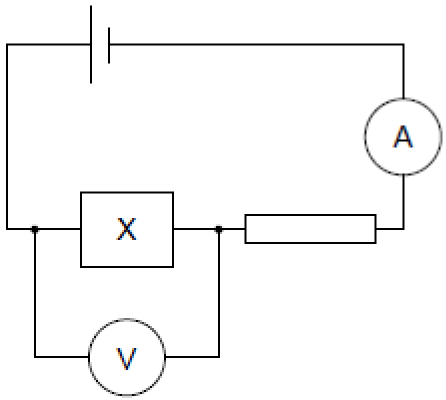
C. $\frac{nV}{e}$

D. $\frac{eV}{n}$ [1]

Markscheme
B

29. [Maximum mark: 1] 19M.1.SL.TZ2.19

The resistance of component X decreases when the intensity of light incident on it increases. X is connected in series with a cell of negligible internal resistance and a resistor of fixed resistance. The ammeter and voltmeter are ideal.



What is the change in the reading on the ammeter and the change in the reading on the voltmeter when the light incident on X is increased?

	Ammeter reading	Voltmeter reading
A.	increases	decreases
B.	increases	increases
C.	decreases	decreases
D.	decreases	increases

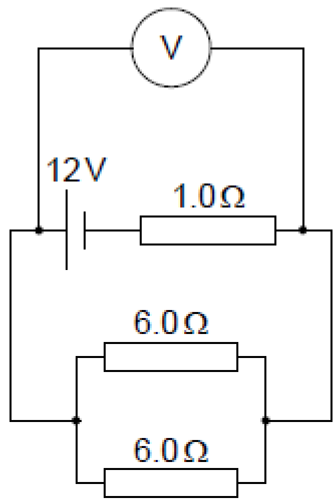
[1]

Markscheme

A

30. [Maximum mark: 1] 19M.1.SL.TZ2.20

Three resistors of resistance 1.0 Ω, 6.0 Ω and 6.0 Ω are connected as shown. The voltmeter is ideal and the cell has an emf of 12 V with negligible internal resistance.



What is the reading on the voltmeter?

A. 3.0 V

B. 4.0 V

C. 8.0 V

D. 9.0 V [1]

Markscheme

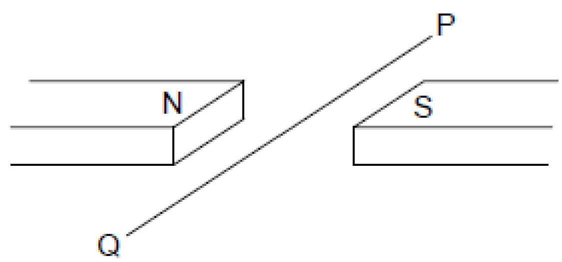
D

Examiners report

Most candidates at both levels gave option A as the correct response instead of D. This would indicate that they have misread the diagram thinking the voltmeter was across the 1.0Ω resistor not the parallel combination.

31. [Maximum mark: 1] 19M.1.SL.TZ2.21

A horizontal wire PQ lies perpendicular to a uniform horizontal magnetic field.



A length of 0.25 m of the wire is subject to a magnetic field strength of 40 mT. A downward magnetic force of 60 mN acts on the wire.

What is the magnitude and direction of the current in the wire?

	Current magnitude / A	Current direction
A.	6.0	P to Q
B.	6.0	Q to P
C.	0.17	Q to P
D.	0.17	P to Q

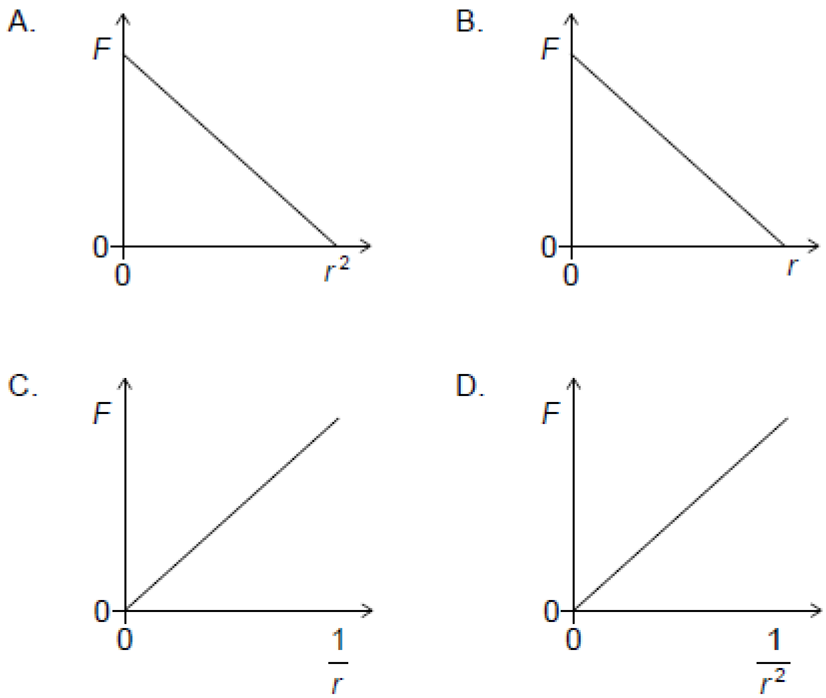
[1]

Markscheme

B

32. [Maximum mark: 1] 19M.1.SL.TZ2.23

Which graph shows the relationship between gravitational force F between two point masses and their separation r?



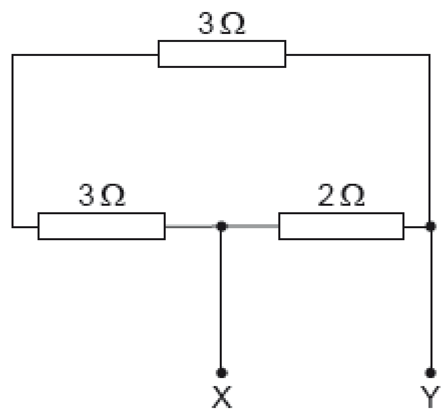
[1]

Markscheme

D

33. [Maximum mark: 1] 18M.1.SL.TZ1.18

Three resistors are connected as shown. What is the value of the total resistance between X and Y?



A. 1.5 Ω

B. 1.9 Ω

C. 6.0 Ω

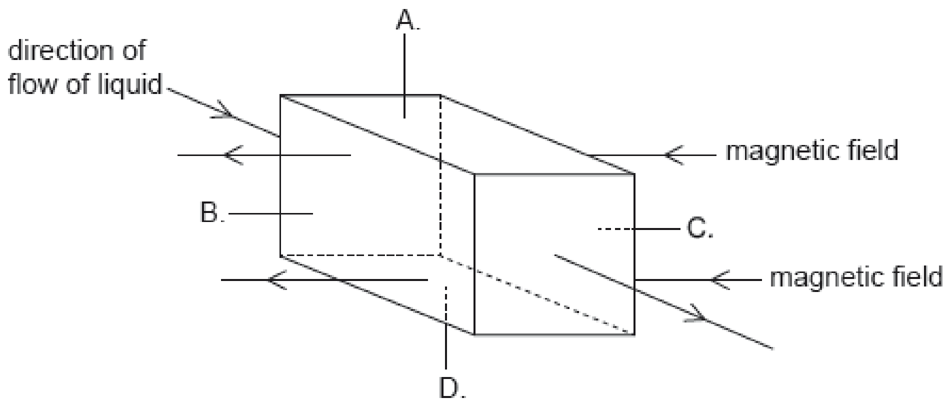
D. 8.0 Ω [1]

Markscheme
A

34. [Maximum mark: 1] 18M.1.SL.TZ1.19

A liquid that contains negative charge carriers is flowing through a square pipe with sides A, B, C and D. A magnetic field acts in the direction shown across the pipe.

On which side of the pipe does negative charge accumulate?

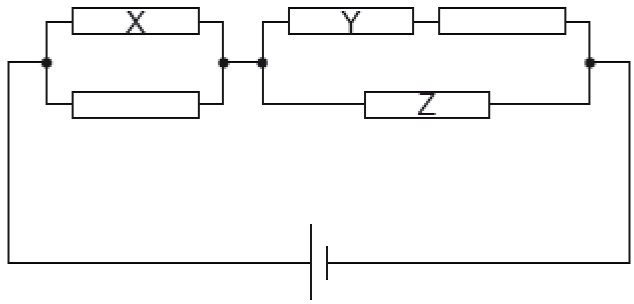


[1]

Markscheme
A

35. [Maximum mark: 1] 18M.1.SL.TZ1.20

Five resistors of equal resistance are connected to a cell as shown.



What is correct about the power dissipated in the resistors?

A. The power dissipated is greatest in resistor X.

B. The power dissipated is greatest in resistor Y.

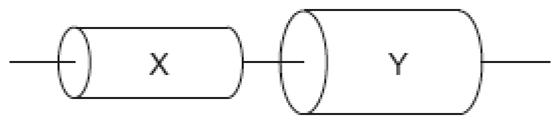
C. The power dissipated is greatest in resistor Z.

D. The power dissipated is the same in all resistors. [1]

Markscheme
C

36. [Maximum mark: 1] 18M.1.SL.TZ1.21

Two resistors X and Y are made of uniform cylinders of the same material. X and Y are connected in series. X and Y are of equal length and the diameter of Y is twice the diameter of X.



The resistance of Y is R.

What is the resistance of this series combination?

A. $\frac{5R}{4}$

B. $\frac{3R}{2}$

C. $3R$

D. $5R$ [1]

Markscheme

D

37. [Maximum mark: 1] 18M.1.SL.TZ1.23

Newton's law of gravitation

A. is equivalent to Newton's second law of motion.

B. explains the origin of gravitation.

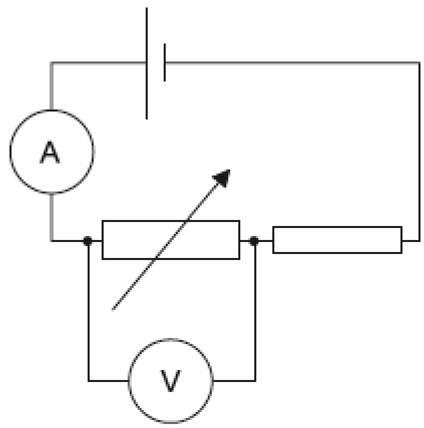
C. is used to make predictions.

D. is not valid in a vacuum. [1]

Markscheme
C

38. [Maximum mark: 1] 18M.1.SL.TZ2.19

A cell with negligible internal resistance is connected as shown. The ammeter and the voltmeter are both ideal.



What changes occur in the ammeter reading and in the voltmeter reading when the resistance of the variable resistor is increased?

	Change in ammeter reading	Change in voltmeter reading
A.	increases	increases
B.	increases	decreases
C.	decreases	increases
D.	decreases	decreases

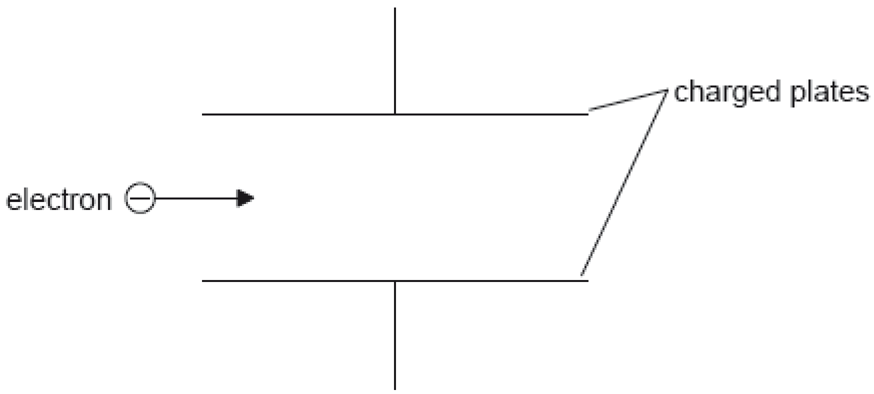
[1]

Markscheme

C

39. [Maximum mark: 1] 18M.1.SL.TZ2.20

An electron enters the region between two charged parallel plates initially moving parallel to the plates.



The electromagnetic force acting on the electron

A. causes the electron to decrease its horizontal speed.

B. causes the electron to increase its horizontal speed.

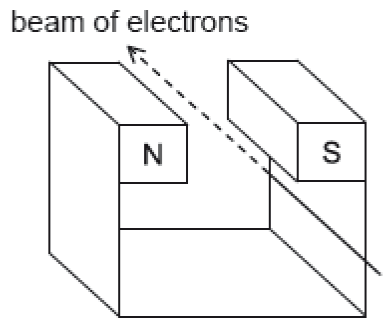
C. is parallel to the field lines and in the opposite direction to them.

D. is perpendicular to the field direction. [1]

Markscheme
C

40. [Maximum mark: 1] 18M.1.SL.TZ2.21

A beam of electrons moves between the poles of a magnet.



What is the direction in which the electrons will be deflected?

A. Downwards

B. Towards the N pole of the magnet

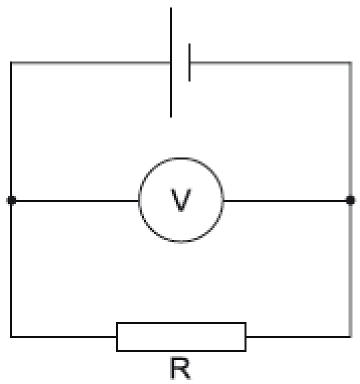
C. Towards the S pole of the magnet

D. Upwards [1]

Markscheme
D

41. [Maximum mark: 1] 18M.1.SL.TZ2.22

A cell has an emf of 4.0 V and an internal resistance of 2.0 Ω. The ideal voltmeter reads 3.2 V.



What is the resistance of R?

A. 0.8 Ω

B. 2.0 Ω

C. 4.0 Ω

D. 8.0 Ω [1]

Markscheme
D